

Bellabeat Capstone Project: Data Cleaning and Preparation Documentation

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Project Objective: To analyze smart device usage data and extract behavioral trends in physical activity, sleep, and health metrics to inform Bellabeat's marketing strategy.

1. Datasets Used

- dailyActivity_merged.csv
- sleepDay_merged.csv
- weightLogInfo_merged.csv
- hourlySteps_merged.csv
- heartrate_seconds_merged.csv
- minuteMETsNarrow_merged.csv

These were selected from the 19 available files to give a broad yet manageable scope.

2. Data Loading

Data was loaded using R functions:

```
daily_activity <- read_csv("dailyActivity_merged.csv")
sleep_day <- read.csv("sleepDay_merged.csv")
weight_info <- read.csv("weightLogInfo_merged.csv")
hourly_steps <- read.csv("hourlySteps_merged.csv")
heart_rate <- read.csv("heartrate_seconds_merged.csv")
mets_data <- read.csv("minuteMETsNarrow_merged.csv")
```

3. Duplicate Removal

Each dataset was checked and cleaned of duplicate rows:

```
daily_activity <- distinct(daily_activity)
sleep_day <- distinct(sleep_day)
weight_info <- distinct(weight_info)
hourly_steps <- distinct(hourly_steps)
```

```
heart_rate <- distinct(heart_rate)
mets_data <- distinct(mets_data)
```

4. Date-Time Formatting

All date/time columns were carefully formatted using **lubridate**:

```
daily_activity$ActivityDate <- mdy(daily_activity$ActivityDate)
sleep_day$SleepDay <- mdy_hms(sleep_day$SleepDay)
sleep_day$SleepDay <- as_date(sleep_day$SleepDay)
weight_info$Date <- mdy_hms(weight_info$Date)
weight_info$Date <- as_date(weight_info$Date)
hourly_steps$ActivityHour <- ymd_hms(hourly_steps$ActivityHour)
heart_rate$Time <- ymd_hms(heart_rate$Time)
mets_data$ActivityMinute <- ymd_hms(mets_data$ActivityMinute)
```

Special care was taken to test the correct formatting by inspecting:

```
head(column_name)
colnames(dataset)
str(dataset)
```

5. Handling Time-Series Granularity

The datasets included second-, minute-, hourly-, and daily-level granularity. These were formatted accordingly, and key datetime variables (hour, minute, etc.) were extracted where needed using:

```
hour(datetime_column)
minute(datetime_column)
date(datetime_column)
```

6. Spreadsheet Preprocessing

In addition to R, significant preprocessing was done using spreadsheets:

- Duplicate check
- Date formatting inconsistencies manually resolved

- Created new columns for weekday extraction and time segmentation
- Applied conditional formatting to identify patterns by user ID
- Pivot tables created for weekday vs. steps, calories, etc.

7. File Consolidation

For merged datasets:

```
activity_sleep <- merge(daily_activity, sleep_day, by.x = c("Id", "ActivityDate"), by.y = c("Id", "SleepDay"))
```

8. Plot Testing and Recovery

- Some charts initially failed to render due to date parsing issues
- Fixed using correct lubridate functions (mdy_hms, ymd_hms)
- Plot chunk alignment was fixed with fig.width and fig.height
- Emojis in Markdown caused LaTeX knitting errors — resolved by switching to HTML output and removing problematic Unicode

9. Knitting & Export

- Final .Rmd successfully knitted to HTML format
- All plots rendered inline
- Document structured with Markdown headings and plot insights

10. Final Deliverables

- R Markdown file (capstone.Rmd) also available as HTML Report (capstone.html) and Pdf document (capstone.pdf)

- PowerPoint summary created with visuals from spreadsheets

Status: Data cleaning and preparation successfully completed. Visualization and reporting phases done

Recommendations are presented in the powerPoint